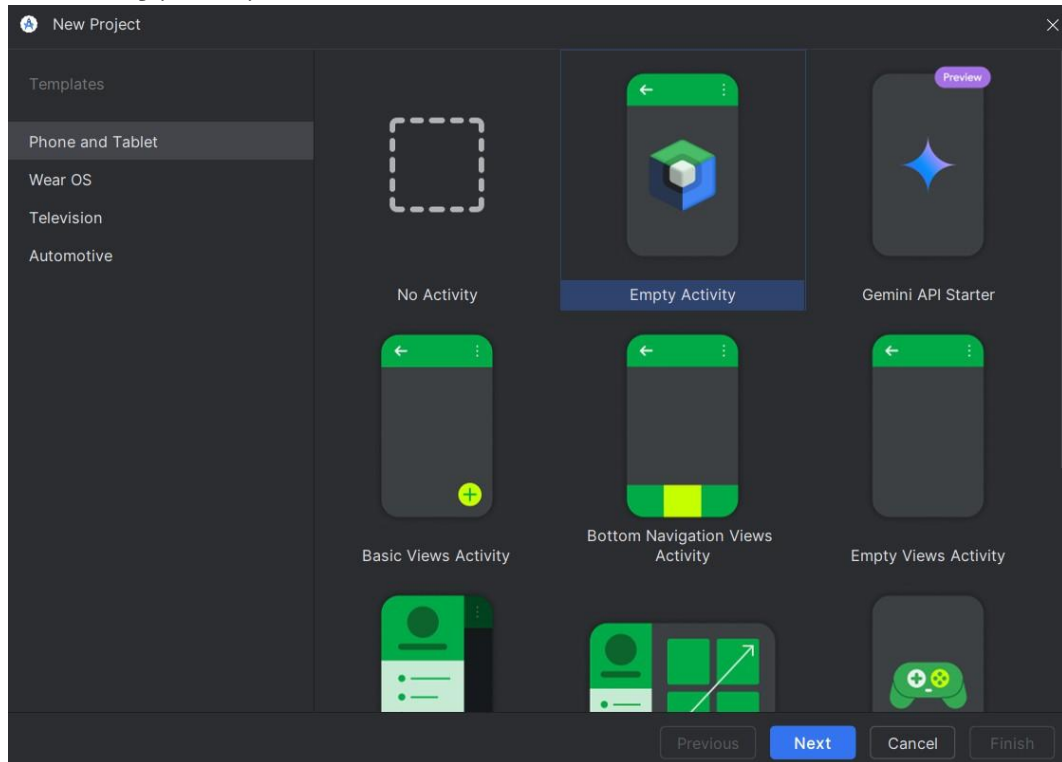


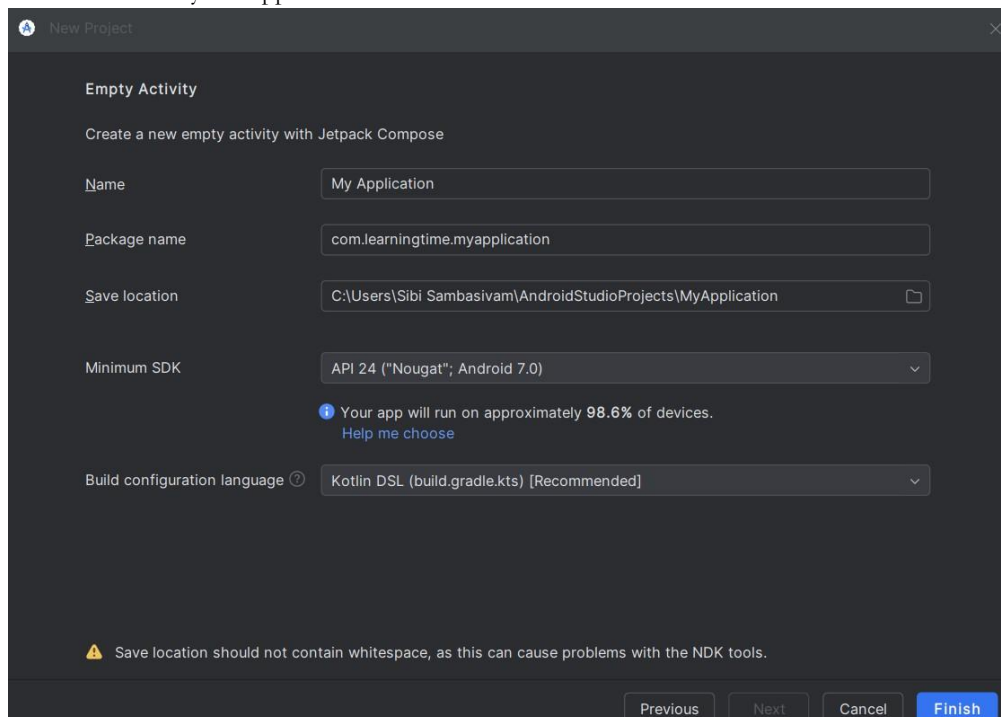
ACTIVITIES AND THEIR LIFECYCLE

ANDROID APPLICATION DEVELOPMENT – 20CYS404

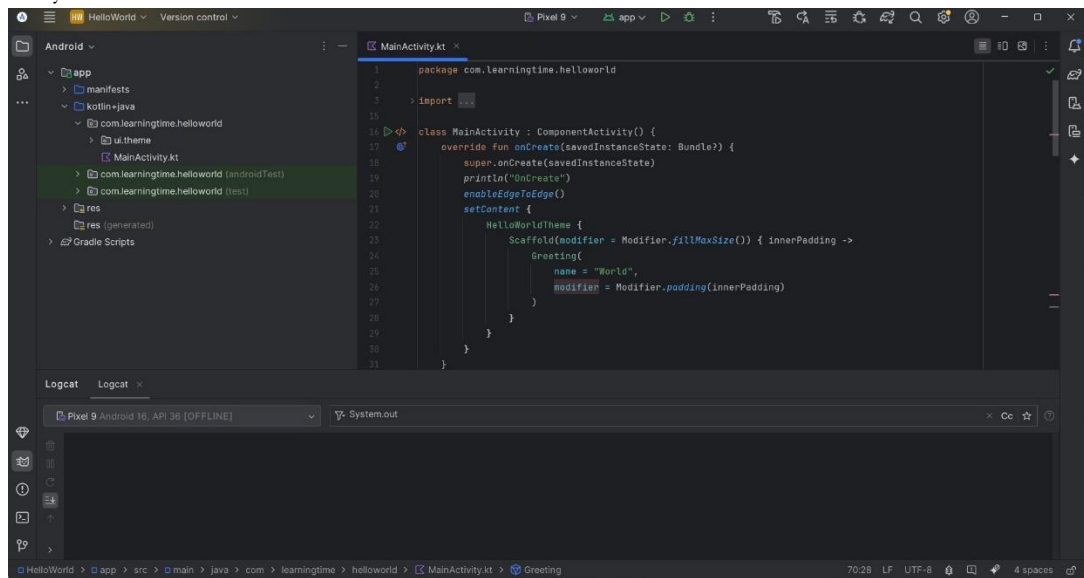
1. Open Android Studio and Create a new Project
2. Click on Empty Activity



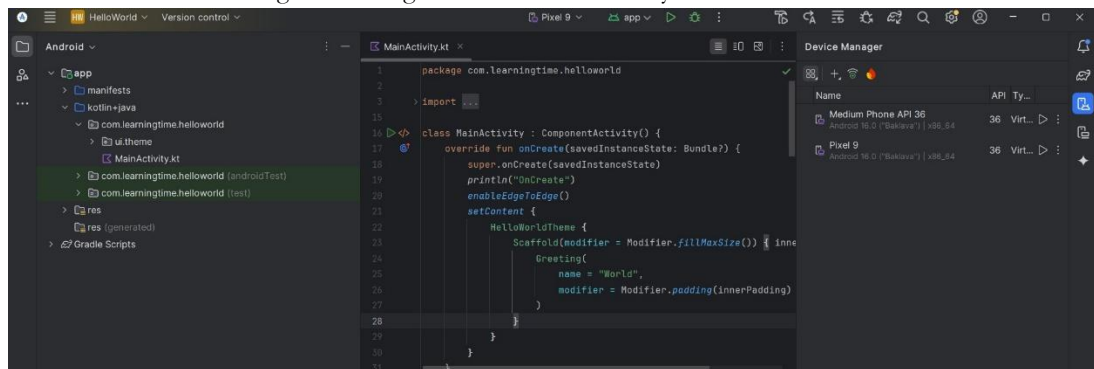
3. Write a name for your Application and then click Finish.



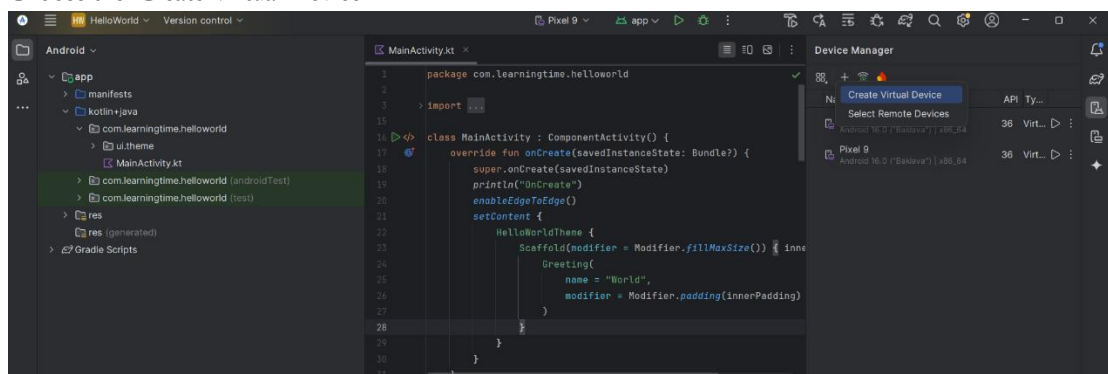
- Wait for the gradle to build your application. After the gradle has finished building, your application will be ready to be executed.



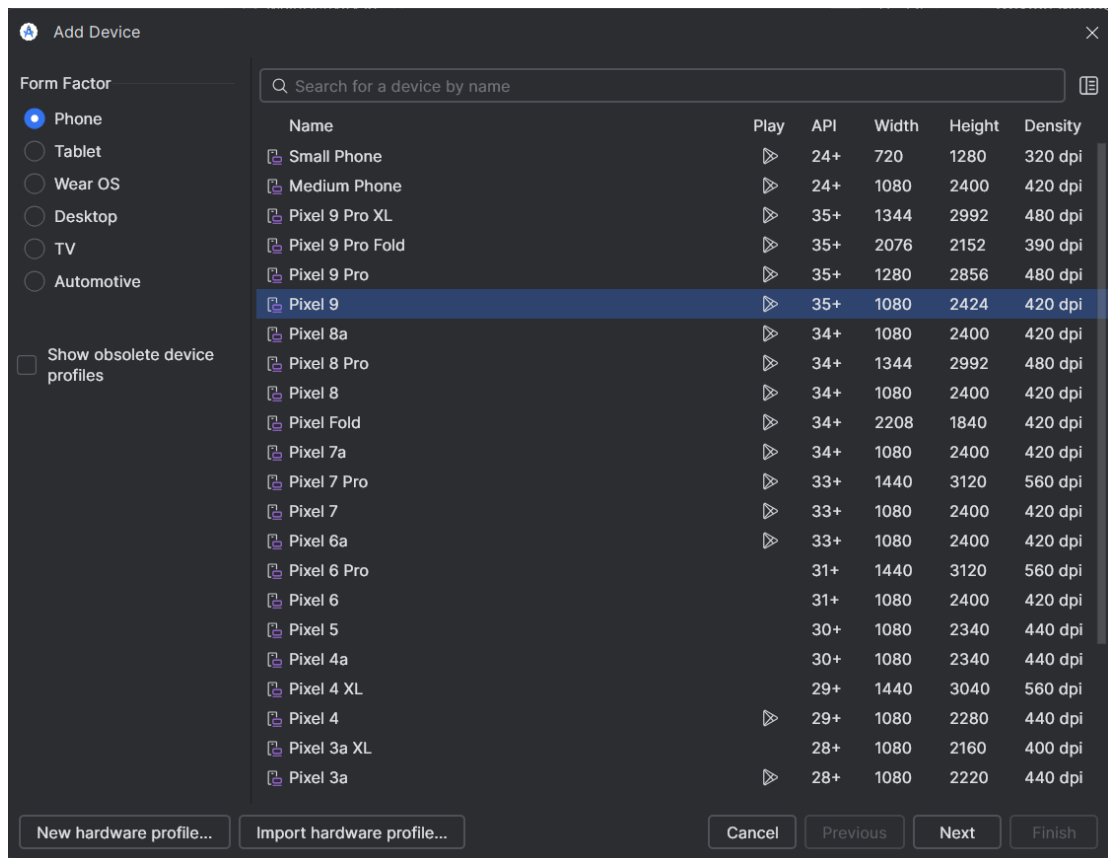
- Click on the Device Manager on the right and click on the '+' symbol.



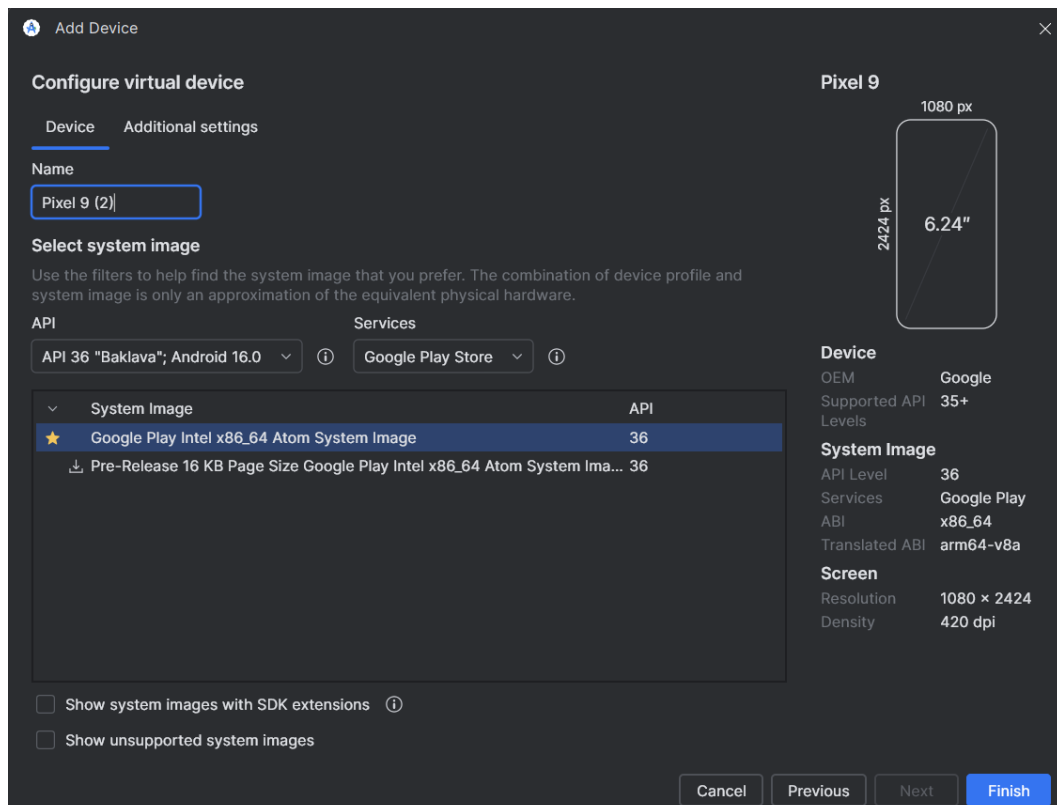
6. Choose the 'Create Virtual Device'.



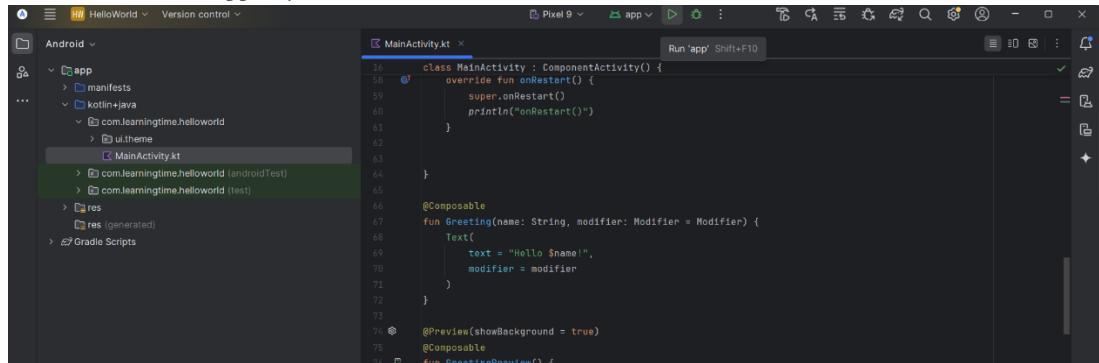
7. In the Add Device menu, choose a phone model with the Google Play symbol available to it and click Next.



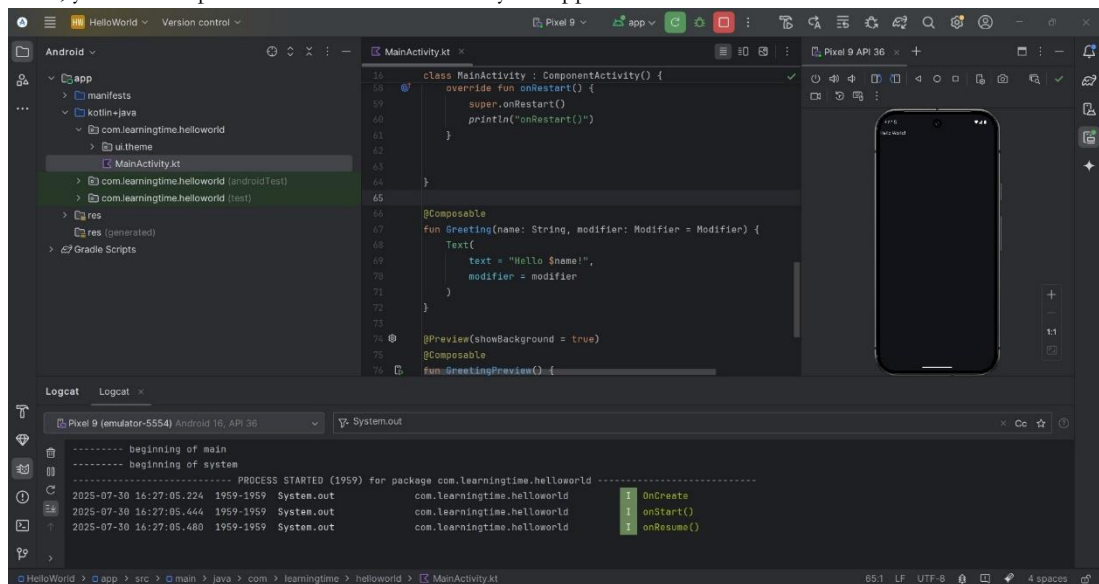
8. In the next window, Click Finish. The device emulator will be downloaded and added to your Android Studio.



9. Click on the “Run ‘App’” symbol above.



10. Now, your virtual phone should switch on and your app will load on the device.



11. After adding the code provided below into your MainActivity.kt file, Open your logcat and in the filter, Type “ System.out”. You should see 3 Activities being executed.

```
package com.learningtime.helloworld
import ...

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        println("OnCreate")
        enableEdgeToEdge()
        setContent {
            HelloWorldTheme {
                Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
                    Greeting(
                        name = "World",
                        modifier = Modifier.padding(innerPadding)
                    )
                }
            }
        }
    }
    override fun onStart() {
```

```

        super.onStart()
        println("onStart()")
    }

    override fun onResume() {
        super.onResume()
        println("onResume()")
    }

    override fun onPause() {
        super.onPause()
        println("onPause()")
    }

    override fun onStop() {
        super.onStop()
        println("onStop()")
    }

    override fun onDestroy() {
        super.onDestroy()
        println("onDestroy()")
    }

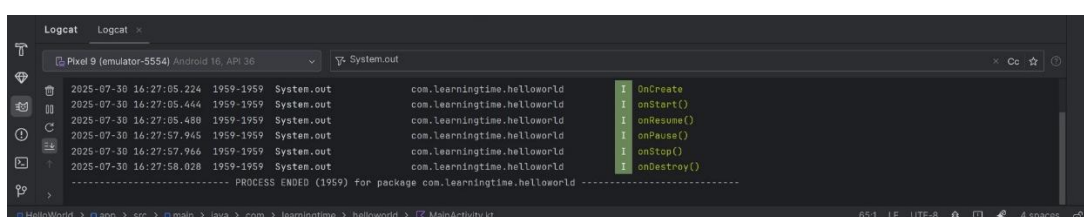
    override fun onRestart() {
        super.onRestart()
        println("onRestart()")
    }
}

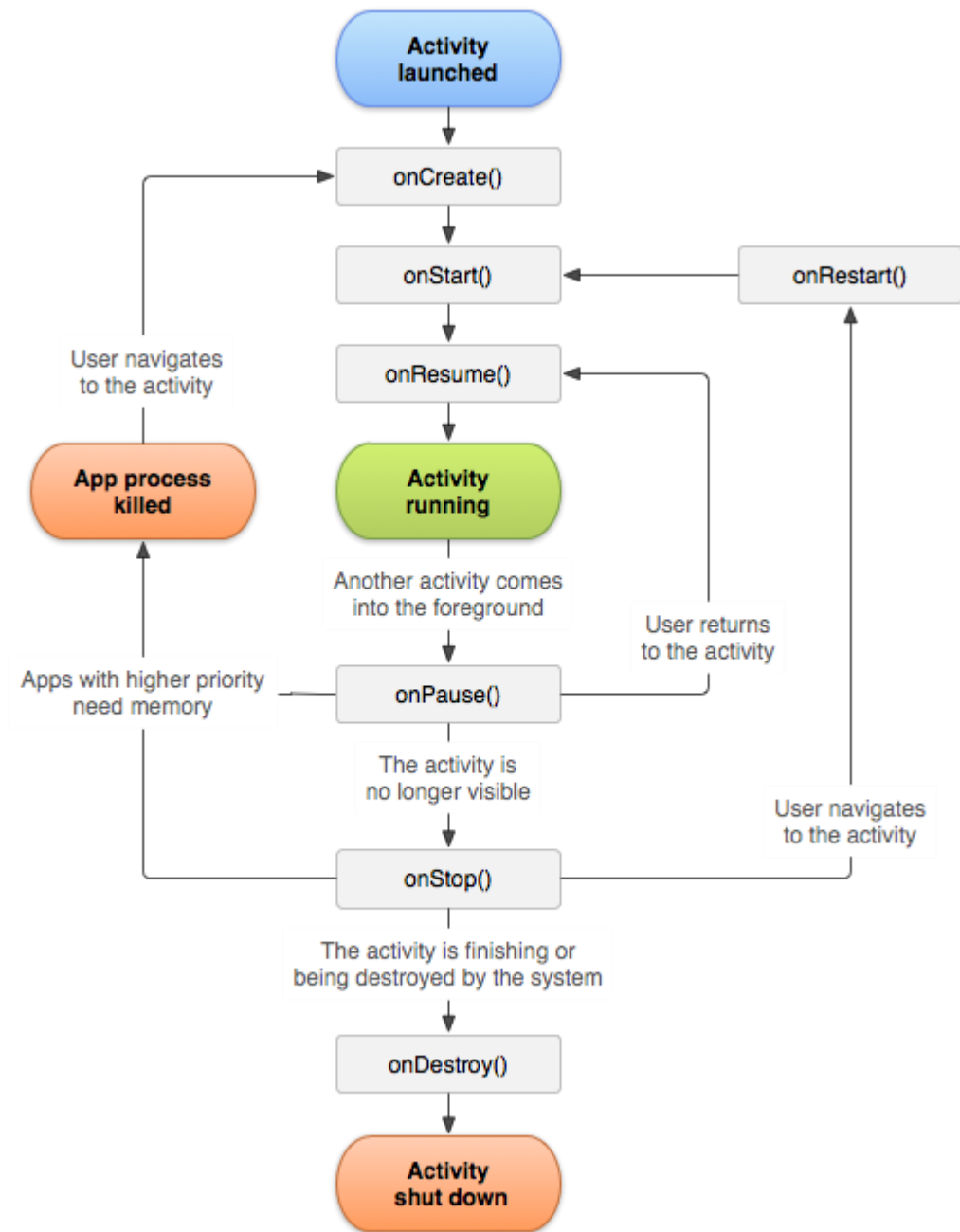
@Composable
fun Greeting(name: String, modifier: Modifier = Modifier) {
    Text(
        text = "Hello $name!",
        modifier = modifier
    )
}

@Preview(showBackground = true)
@Composable
fun GreetingPreview() {
    HelloWorldTheme {
        Greeting("Android")
    }
}

```

12. Interact with the app (let it run in the background, close it and open it again), and you should see the working of the activity lifecycle in the logcat below.





What is an Activity in Android?

An Activity is one of the fundamental building blocks of Android apps. It represents a single screen with a user interface, similar to a window in desktop applications or a page in a web app.

Key Characteristics of Activities:

- **Single Screen Unit:** Each activity manages its own window for drawing UI
- **Independent Components:** Activities are loosely coupled - one activity doesn't directly communicate with another

- **Task Participation:** Activities can be combined to form cohesive user experiences (called "tasks")
- **Entry Points:** Activities are declared in the AndroidManifest.xml as potential entry points to your app

Common Scenarios and Lifecycle Flow

Normal Startup

`onCreate()` → `onStart()` → `onResume()`

User presses Back button

`onPause()` → `onStop()` → `onDestroy()`

User opens another app

`onPause()` → `onStop()`

User returns to your app

`onRestart()` → `onStart()` → `onResume()`

Screen rotation

`onPause()` → `onStop()` → `onDestroy()` →
`onCreate()` → `onStart()` → `onResume()`

Low memory scenario

App process might be killed after `onStop()`,
requiring state restoration in `onCreate()`